

Information Sheet 1: Introduction to Environmental Modelling

EART22001 / EART60071

1 Why start with ordinary differential equations?

We will begin with models that solve *ordinary differential equations* (ODEs). An ODE describes how one or more quantities change, often with time. You may have encountered ODEs previously, but no previous experience is assumed here: the examples below introduce the ideas needed for the modelling work in these modules.

A simple example is

$$\frac{dy}{dt} = -\lambda y,$$

with the initial condition $y(0) = N_0$. Separating variables and integrating gives

$$y = N_0 \exp(-\lambda t),$$

which you may recognise as radioactive decay.

The important modelling idea is not the particular solution. We start from a rule for the *rate of change*, provide an initial state, and calculate how the system evolves. Many environmental problems can be approached in this way.

Some differential equations can be solved exactly; many cannot. Numerical methods approximate their solutions on a computer. A simple example is the forward Euler method. More accurate algorithms are available, and in these modules the emphasis is on using, interpreting and evaluating models rather than deriving every numerical algorithm. For further detail, see Hoffman (1992).

2 What counts as an environmental model?

There is no single kind of environmental model. Across EART22001 and EART60071 you will encounter several useful families of model.

Type	What it means in these modules
ODE models	Models in which an initial state evolves with time. Examples include the evolution of quantities describing environmental parcels, particles, cloud droplets, chemical species and other dynamical systems.
Empirical models	Models that are guided by physical ideas but contain parameters obtained from observations. Gaussian plume modelling is one example used in air-quality work.
Reduced-dimensional models	Deliberately simplified models that represent only the dimensions needed for a particular question. A single-column atmospheric model, for example, represents vertical structure without explicitly modelling the full three-dimensional atmosphere.
Fluid-dynamical models	Models based on equations governing the motion of fluids. These ideas underpin models of the atmosphere, oceans and many other environmental systems.

A model is therefore not automatically “good” because it is complicated. A useful model is one whose assumptions and level of complexity are appropriate to the question being asked.

3 How we will work

For most practicals the numerical model will run on a **Linux server**. Your Windows PC or Mac is the *local* computer. The server is the *remote* computer.

The basic pattern is:

1. connect from your computer to the server using **SSH**;
2. obtain or update the practical files, often using **Git**;
3. edit input or configuration files and run the model on the server;
4. inspect the output;
5. transfer results back to your own computer when needed;
6. analyse, plot and interpret the results.

Do not worry if the command line is unfamiliar. The first practical is designed to teach the small set of commands you need. Keep the separate *Tools of the Trade* guide nearby: it is a reference document, not something you need to memorise before class.

4 A few terms you will hear

Term	Meaning
Terminal	A text-based window in which you type commands. On macOS use Terminal; on Windows use PowerShell, Command Prompt or Windows Terminal.
Linux	The operating system used by the course modelling server.
SSH	A secure way to log in to the remote server and run commands there.
SFTP / SCP	Secure ways to move files between the server and your own computer.
Computer code	Instructions written in a programming language. You will use both interpreted and compiled programs during the modelling practicals.
Git	A version-control system used to keep track of changes to files and code.
GitHub	An online service that hosts Git repositories. We will often obtain practical code from GitHub.
Docker	An optional local Linux environment for practising the modelling workflow off campus. It is not required for the normal on-campus practical workflow.

5 Before the first practical

Please do the following before class:

- ☐ Read this information sheet.
- ☐ Locate the separate *Tools of the Trade* guide on Canvas.
- ☐ Locate your **course server username and password** on Canvas. These are not your central University username and password.
- ☐ Make sure you know how to open PowerShell/Windows Terminal on Windows, or Terminal on macOS.

You do **not** need to install Docker before the first on-campus practical.

Reference: Hoffman, J. D. (1992), *Numerical Methods for Engineers and Scientists*, McGraw-Hill.